Adaptive AI Technical Interviewer

Mid-term report

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Code: `https://github.com/Srinjoy003/LMA-RAG`

Abstract

We are building an interviewer for junior technical screening in Python, SQL and data structures. It keeps an estimate of the candidate's level in each skill, picks each question to reduce its uncertainty, stops when it is confident, and works out *why* a wrong answer is wrong. It recommends fit, not fit or human review and never makes the hiring decision. For this report we ran the full system (A2) and a fixed 12-question interview (B1) on the same 24 simulated candidates, with Gemini grading and a different model playing the candidate. The adaptive system placed 72.2% of skill levels correctly against 55.6% for the fixed interview, a gain of 16.7 points (95% interval 4 to 29 points). But it also asked more: 21.0 exchanges against 12.0, mostly follow-ups after wrong answers. So the live gain mixes adaptive question choice with extra questions. In simulated runs with a perfect grader, adaptive choice at exactly 12 questions beat the fixed list by about 6 points; how much of the live gain it explains is not yet measured. It confirmed 19 of the 35 planted misconceptions, which the fixed interview cannot do, and rarely named one on a correctly answered reply (3.3%). However, only 18% of the misconceptions it confirmed were planted ones; most of the rest came from wrong answers, and nearly all were confirmed from a single question. Our proposal promised 25% *fewer* exchanges; we do not meet that, and we explain why we chose accuracy over length.

1 What we set out to do

A screening interview has to judge several skills with a handful of questions. Fixed question lists waste questions on skills that are already clear and leave others under-tested. LLM interviewers make screening cheap, but they decide what to ask and when to stop from a prompt alone, with no record of how sure they are. Adaptive testing solves this for multiple-choice tests by asking the most informative question and stopping once the estimate is precise [9], and TestAgent [8] brings the idea to conversational assessments, reporting about 20% fewer questions. We apply it to open answers across several skills, and add a model of the candidate's misconceptions. Our proposal asked two questions:

RQ1 Can uncertainty-driven questioning with early stopping match a fixed 12-question interview's accuracy with fewer exchanges?

RQ2 Does modelling what the candidate *believes* (their misconceptions) find those misconceptions better than a scorecard that only tracks skill?

All four parts of the system are now built and tested: the question bank, the interview engine, the grader and diagnoser, and the evaluation harness with simulated candidates. This report covers the first live experiment. It compares B1 and A2. The ablations that answer RQ2 directly (A4, skill-only scorecard; A5, job-blind retrieval) are built but not yet run live.

Main findings so far.

- The adaptive system judges levels clearly better than the fixed interview (72.2% against 55.6%). Observed verdict accuracy was similar, but with 24 candidates we can show neither that A2's verdicts are better nor that they are no worse.
- It is 75% longer, so part of the accuracy gain comes from asking more. In controlled runs at equal length, adaptive question choice alone still beat the fixed list (Section 5.2).
- It finds about half of the planted misconceptions (54%) and rarely names one on a correct answer, but its confirmed misconceptions are mostly not planted ones (precision 18%). Section 7 explains where they come from.

2 Changes since the proposal

Part	Proposal	Now
Question bank	90 questions	135 questions (15 per skill and level), written for 3 job descriptions
Question choice	by skill and level	by skill and level, preferring topics the job needs (revision plan, change 1)
Diagnosis	weighted hypotheses	one structured call with fixed causes, run 3 times; votes give the weights (change 2)
Grader	fine-tuned 7–8B model	prompted Gemini model; fine-tuning moved to stretch (change 3)
Systems run	B1, A1–A4 on 60 candidates	B1 and A2 on 24 candidates (two balanced blocks of 12)
Stopping target	25% fewer exchanges	kept the threshold that gave the best accuracy; length is reported, not tuned for

The last row is a change of objective. RQ1 asked for B1's accuracy with fewer exchanges. On our tuning runs (Section 5.1) a stricter stopping threshold met the letter of that target: 6.8 exchanges, 44% fewer than B1, at 65.4% accuracy against B1's 70.0%, within the 5-point margin but less accurate than the fixed interview. We chose instead to keep the threshold that gave clearly better accuracy, and to report the length honestly. This puts accuracy ahead of length, which differs from what RQ1 originally asked.

3 The system

Figure 1 shows the loop. The LLM is used only where language is needed. Everything that controls the interview (which question, when to stop, the verdict rule) is plain code, so the candidate cannot change the procedure by asking. The grades and diagnoses still come from an LLM, so a persuasive answer could in principle sway them; our safety check (two manipulation attempts, Section 6) is too small to rule that out.

Scorecard. Each skill has an ability value θ on a logit scale. Questions come in three levels with difficulty $b = -1, 0, +1$. An answer earns 0, 1 or 2 marks, and we model the chance of each mark with the Partial Credit Model [5]:

$$P(X = x \mid \theta, b) \;\propto\; \exp\Big(\sum_{k \le x} (\theta - \delta_k)\Big), \qquad \delta_1 = b - 0.5,\ \delta_2 = b + 0.5.$$

We start from a normal prior (mean 0, standard deviation 1) and update it on a grid after every graded question. The estimate is the posterior mean (EAP, [2]) and the uncertainty is the posterior standard deviation (SE). Each answer is graded three times; if the three marks disagree, the answer counts as weaker evidence (2 of 3 agreeing gives weight 0.67).

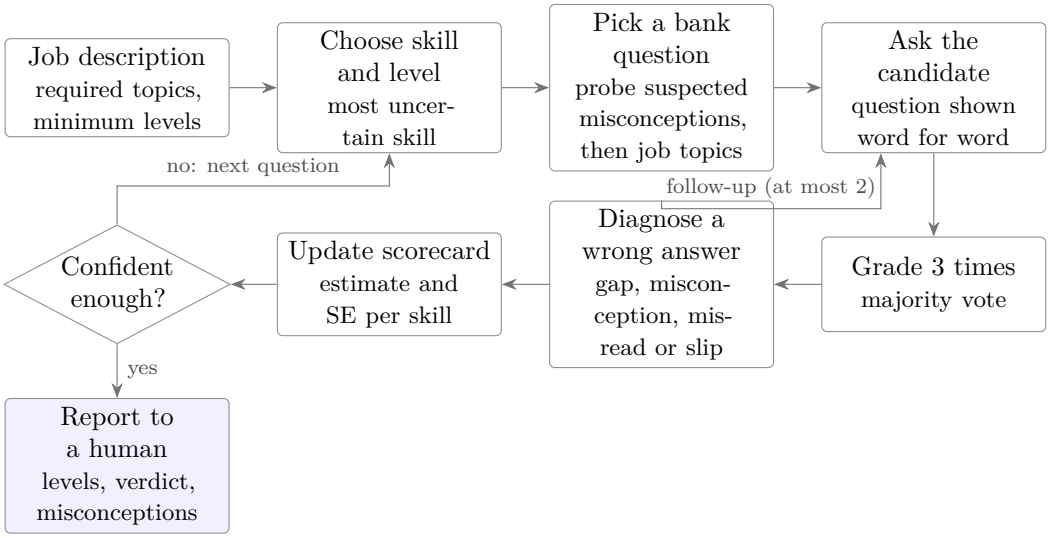


Figure 1: One interview. The LLM grades, diagnoses and writes follow-up questions; code makes every decision (which question, when to stop, the verdict).

From ability to levels. The continuous estimate is turned into a level 0–3 with three cut points on the same scale (Section 5.1 explains how they were set):

Ability θ	below -1.25	-1.25 to -0.25	-0.25 to 0.75	0.75 and above
Reported level	0	1	2	3

The same cut points link a job's minimum level to the ability scale: a minimum of level m means θ must be at or above the lower edge of level m (-1.25 for level 1, -0.25 for level 2, 0.75 for level 3; a minimum of 0 is always met).

Choosing the next question. The next skill is the one with the highest SE. The level is the one whose difficulty is closest to that skill's estimate. Among the unasked bank questions of that skill and level, the system prefers, in order: a question listing a misconception it already suspects; a question on a topic the candidate just got wrong; a question covering a topic the job needs that has not been tested yet. Remaining ties are broken by a seeded random choice, so the same answers always lead to the same questions.

Stopping. The interview stops when every skill has at least two answers and an SE below 0.6, or after 15 main questions.

Grading, diagnosis and follow-ups. Each question goes through these steps.

1. The grader marks the main answer three times (0, 1 or 2). The most common mark is the *provisional* grade; if all three marks differ, the middle one is used.
2. If the provisional grade is below 2, the diagnoser is called three times. Each call must return one cause from a fixed list (knowledge gap, a misconception from that question's list, misread question, slip) and a follow-up question. The most common (cause, misconception) pair wins. If all three calls disagree, the first one wins with a vote share of 1/3; this happened in 2 of the 243 diagnoses that named a misconception.
3. A follow-up is asked unless the cause is a plain gap or the misconception is already confirmed, at most two per question. Each follow-up reply is diagnosed the same way.
4. The grader then marks the *whole exchange* (main answer and follow-ups) three times. This final grade, not the provisional one, updates the scorecard, and the share of the three marks that

agree with it is the evidence weight. When a question had no follow-ups, the whole exchange is just the main answer, so its provisional grade is reused and no new calls are made.

A misconception counts as *confirmed* once it has won the diagnosis vote on two replies. The two replies may be a main answer and its own follow-up; Section 7 shows why this matters. One winning reply only makes it "suspected".

Verdict. For each skill the system computes, from the posterior, the probability that θ is at or above the cut point of the job's minimum level. Above 0.8 the skill meets the minimum, below 0.2 it does not, and in between it is uncertain; a skill with fewer than two graded answers is always uncertain. So a skill can be reported at level 2 and still be "uncertain" against a level-2 minimum if the estimate sits close to the cut. The verdict is *not fit* if any skill is clearly below, *fit* if all clearly meet, and *needs human review* otherwise.

The baseline (B1). B1 asks four questions per skill, at levels 1, 2, 2 and 3, rotating through the skills. It uses the same bank, the same job-aware question ranking and the same grader, but never adapts, never stops early and asks no follow-ups. So any difference between B1 and A2 comes from how questions are chosen, when the interview stops, and the diagnosis.

Agent version. The proposal asked for tool calling. The engine also runs with an LLM agent in charge, calling tools such as `search_question_bank`, `update_scorecard` and `end_interview`. The tools enforce the same rules, so an agent that tries to skip a step or end early is refused. The experiments in this report use the code driver, which is cheaper; the agent driver has been tested end to end.

4 Data

Question bank. 135 questions: 3 skills × 3 levels × 15. Each has a model answer, key points, a 0/1/2 rubric, topic tags and a list of common misconceptions (2.3 on average). There are two kinds: *explain* questions and *read-code* questions that ask what a snippet prints and why. Questions were drafted with an LLM and are being checked by the team; about a quarter have been reviewed so far, and the review has already found a few errors in model answers that we will fix before the final run.

Misconceptions. 94 misconceptions across the three skills. Where a published catalogue exists we adapted from it: Progmiscon for Python [3], a think-aloud study of SQL novices [6], concept-inventory work on data structures [4] and the McMining benchmark [1]. The other 73 were written for this project.

Jobs. 3 job descriptions (junior data analyst, backend engineer, data engineer), each with a minimum level per skill and 13 to 17 required topics. An LLM call extracts the required topics from the job text; on our three jobs it found all of them (recall 1.00, precision 0.96).

Simulated candidates. Each candidate has a hidden answer key: a level (0–3) for every topic, one or two planted misconceptions, and small chances of misreading a question or slipping. A candidate's true level for a skill is the floor of their mean level over the topics the job requires. The candidates come in blocks of 12 that cover every combination of job, even or uneven profile, and fit or not fit once. This run used the first two blocks (24 candidates, 35 planted misconceptions, 2 manipulation attempts).

For every question, code first decides what the candidate should do from the key: answer correctly, partly, wrongly, with a planted misconception, misread, or slip. A second LLM then writes the answer following that plan. It never sees the model answer and is told which points to

get right or wrong. Because the plan is recorded, we know the intended mark for every reply. We use a different model family for the candidate than for the grader, so the grader cannot simply recognise its own writing.

5 Experimental setup

Models. Grading, diagnosis and topic extraction use `gemini-3.5-flash-lite`. The candidate is `gpt-oss-120b` served by Groq. We picked these from small tests: a local Llama 3 model found only 2 of 6 misconceptions as a grader, and the smaller `gpt-oss-20b` followed its answer plans 77% of the time against 93% for the 120b model.

Runs. Each of the 24 candidates was interviewed once by B1 and once by A2, 48 interviews in total. The interviewer-side calls per interview (Gemini; the candidate's own Groq calls are not counted) follow from the procedure above: 3 grading calls per main question, 3 more for each question that had follow-ups, 3 diagnosis calls per diagnosed reply, and 1 call to extract the job's topics. That gives 37 calls for a B1 interview (12 questions) and 104 on average for A2; the recorded counts match this formula for all 48 interviews. Almost all of it ran on free API tiers spread across four accounts; the last part used a paid Gemini key and cost about $1. The candidate model ran on Groq's free tier throughout, so this is the whole cost.

Metrics. *Skill accuracy* is the share of (candidate, skill) pairs where the reported level equals the true level. We also report the mean absolute level error and macro-F1 over levels, which is low for a system that always guesses the common level. *Exchanges* are main questions plus follow-ups. *Fit accuracy* is measured on the interviews that reached a fit or not-fit verdict; "needs review" is counted separately, not as an error. *Misconception recall* is the share of planted misconceptions that were confirmed, and *precision* the share of confirmed misconceptions that were planted. The *false-alarm rate* is the share of replies planned as correct on which any misconception was named.

Uncertainty. With 24 candidates the numbers move a lot from sample to sample, so every row of the results table comes with a 95% bootstrap interval: we redraw the candidates with replacement 2,000 times, using the same draw for both systems because they interviewed the same people. For metrics both systems have, the interval is for the difference between them; for A2-only metrics it is for A2's value.

5.1 Tuning before the live run

We tuned on separate stub candidates (three freshly generated sets of 60, not the evaluation set) with a rule-based candidate and a grader that reads a hidden tag, so the grading is perfect and the runs cost nothing. Two things came out of this.

Level cut points. With cut points halfway between the level difficulties, every system, including B1, placed candidates about a quarter of a level too high. The reason is how a level is defined: a level-2 candidate answers level-2 questions fully and gets partial credit on level-3 questions about half the time, which puts their estimate about half a step *above* the level-2 difficulty. Moving the cut points up by 0.25 (to -1.25, -0.25, 0.75) removed the bias and raised accuracy for every system.

Stopping threshold. Figure 2 shows accuracy against length for several thresholds. Stopping at SE below 0.75 gives only 6.8 exchanges, but accuracy drops to 65.4%, below B1's 70.0%. At 0.7 it is 10.1 exchanges and 71%, barely above B1. At 0.6 the system reaches 79% with 15.2 exchanges. Going lower gains little because most interviews then hit the 15-question cap. We kept 0.6.

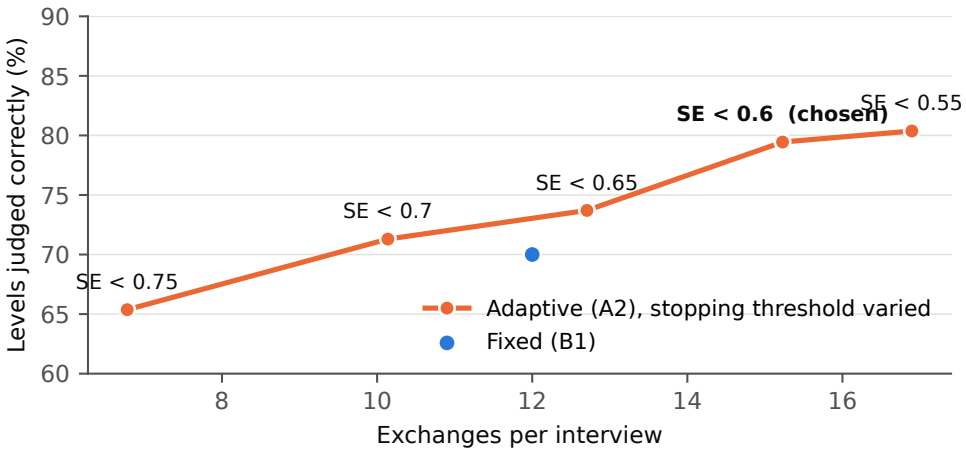


Figure 2: Tuning the stopping threshold on stub candidates with a perfect grader (180 candidates per point). A shorter interview is possible, but only by giving up the accuracy gain over B1.

5.2 Is the gain just more questions?

A2 asks more than B1, so a fair question is whether B1 would do as well with more questions, or whether adaptive choice helps at the same length. We checked this on the same stub candidates with three extra configurations (Table 1).

Configuration	Main questions	Exchanges	Levels right
B1: fixed list, 12 questions	12	12.0	70.0%
B1 stretched: fixed list, 15 questions (levels 1, 2, 2, 3, 3)	15	15.0	71.3%
Adaptive choice, exactly 12 questions, no follow-ups	12	12.0	75.9%
Adaptive choice, exactly 12 questions, with follow-ups (A1)	12	13.5	77.4%
A2 as run (adaptive, early stopping, follow-ups)	up to 15	15.2	79.4%

Table 1: Equal-length controls on stub candidates (180 per row, perfect grader).

Three extra fixed questions barely help B1. At exactly 12 questions with no follow-ups, the same length as B1, adaptive choice gets about 6 points more. Follow-ups and the longer, adaptive stopping add the rest. These are single simulated runs with a perfect grader, so we read them as a sign that adaptivity helps at equal length, not as a measure of how much it contributes to the live result: the live grader makes mistakes and the live gain is larger. Running these controls with the live models is the way to split the live gain, and it is on our list (Section 9).

6 Results

Table 2 has the main numbers. We go through them in turn.

Skill levels. A2 judged 72.2% of skill levels correctly and B1 55.6% (Figure 3). The interval for the difference, 4 to 29 points, stays above zero even with this small sample, so the gain is unlikely to be luck. A2 is better on all three skills. Because A2 also asked more, this live gain is not all due to adaptivity. The simulated equal-length controls (Section 5.2) suggest adaptivity contributes, but we cannot yet say how much of the live gain it explains. The macro-F1 difference points the same way but its interval includes zero. Figure 4 shows where the errors are. B1 mostly errs upwards (25 levels too high against 7 too low). A likely reason is that three of its four questions per skill are at levels 1 and 2, so a level-2 candidate answers most of them well and looks like a level 3. A2 errs upwards much less (15 against 5), because it keeps asking at the level where the candidate's answers start to fail.

	Fixed (B1)	Adaptive (A2)	A2 − B1 [95% interval]
Levels judged correctly	55.6%	72.2%	+16.7 points [+4, +29]
Mean level error (lower is better)	0.46	0.28	−0.18 [−0.31, −0.06]
Macro-F1 over levels	0.46	0.65	+0.19 [−0.04, +0.42]
Exchanges per interview	12.0	21.0	+9.0 [+7.8, +10.4]
Fit verdict correct (when decided)	90%	92%	+2 points [−22, +27]
Sent to human review	58%	50%	−8 points [−29, +17]
Wrong fit or not-fit verdicts	1 of 24	1 of 24	0 points [−12, +12]
Grades matching the planned mark	88%	84%	−4 points [−9, +1]
			A2 [95% interval]
Misconception recall (19 of 35 planted)	–	54%	[36%, 73%]
Misconception precision (19 of 103 confirmed)	–	18%	[12%, 25%]
False alarms on correct replies	–	3.3%	[0.6%, 6.7%]

Table 2: Main results on 24 candidates. A2 asked 13.8 main questions and 7.2 follow-ups per interview.

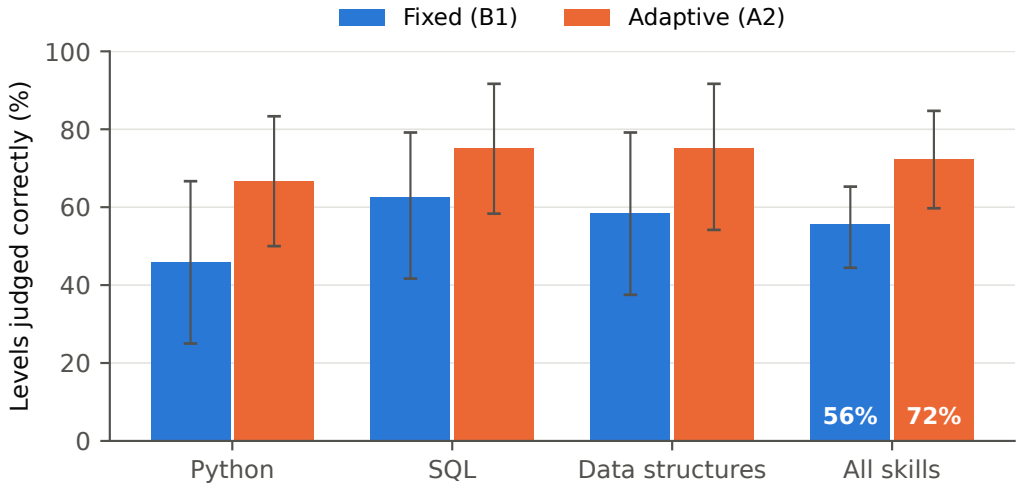


Figure 3: Share of skill levels judged correctly, with 95% bootstrap intervals.

Verdicts. Both systems gave one wrong verdict, and in both cases it passed a candidate who should not have passed (Figure 5). A2's error was a candidate whose Python level is 1, judged as 2. A2 reached a right verdict slightly more often (11 against 9) and sent fewer candidates to a human, but the intervals in Table 2 are wide and include no difference, so we do not claim A2 decides better. Both still send about half the candidates to review. This is the fit rule being careful: it needs 80% confidence either way, and 24 candidates with two to five questions per skill often fall short of that. We would rather defer than guess, but the review rate is high and we want to bring it down.

Length. A2 averaged 21.0 exchanges against B1's 12.0 (Figure 6). The extra +9.0 exchanges are 1.8 more main questions (13.8 against 12) and 7.2 follow-ups, so follow-ups make up 80% of the increase. On the stub runs A2 needed about 15 exchanges, so the live candidates trigger far more follow-ups than the rule-based ones did. The next paragraph explains why. Six A2 interviews hit the 15-question cap.

Misconceptions. A2 confirmed 19 of the 35 planted misconceptions (recall 54%). It also confirmed many that were not planted: 103 in total, so only 18% of its confirmed misconceptions were planted ones (Figure 7). As a misconception report for a hiring manager, this is the weakest part of the system today. Almost all of the extra ones (79 of 84) came from replies the candidate was

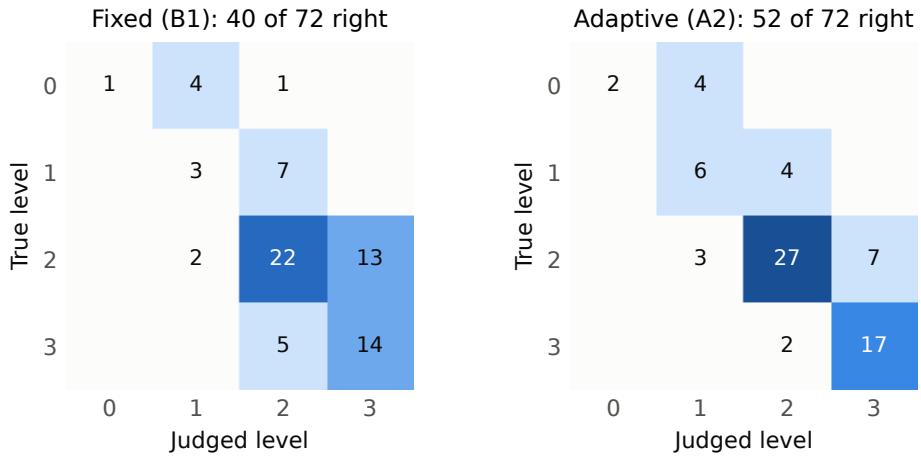


Figure 4: True level against judged level, all skills pooled. Counts on the diagonal are correct.

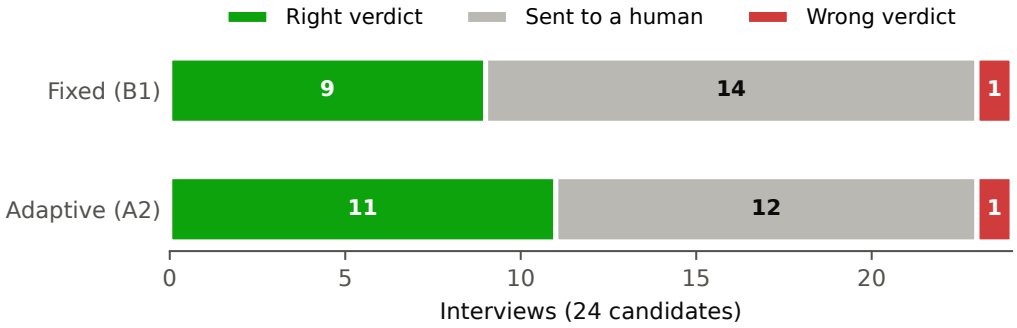


Figure 5: Fit or not-fit verdicts for the 24 candidates.

told to answer wrongly or partly. On replies planned as correct, the system named a misconception only 3.3% of the time, well under the 15% limit in our proposal. For comparison, Oli et al. found that LLMs reviewing complete and correct student code explanations reported problems that were not there 27% of the time [7]. Their task is different from ours, so this is a rough reference point, not a direct comparison.

An interview from the inside. Figure 8 shows one A2 interview. The system asks about the most uncertain skill each time, so the three uncertainty curves come down together, and it stops after 13 questions once all three are below 0.6. Here is a short exchange from another interview, where a planted misconception was confirmed:

> **Question** (SQL, level 2): a query lists `departments` and `employees` in `FROM` with no join condition. What does it return?
> **Candidate:** "...because the two tables are listed together in the FROM clause, the database automatically matches each employee to its department via `dept_id`."
> **Follow-up:** If you have 3 departments and 4 employees, how many rows would `FROM departments, employees` give without a WHERE clause?
> **Candidate:** "One row per employee, so four rows."
> **Result:** confirmed: *believes tables in FROM are matched on their shared key automatically* (the true answer is 12 rows, every combination).

Grading and safety. For each main question, we compare its final grade (which takes any follow-ups into account) with the mark the candidate's last reply was planned to earn. They agree on 84% of A2's main questions and 88% of B1's. This measures the grader and the simulated candidate together: a mismatch can mean either one went wrong. Two candidates tried to manipulate the

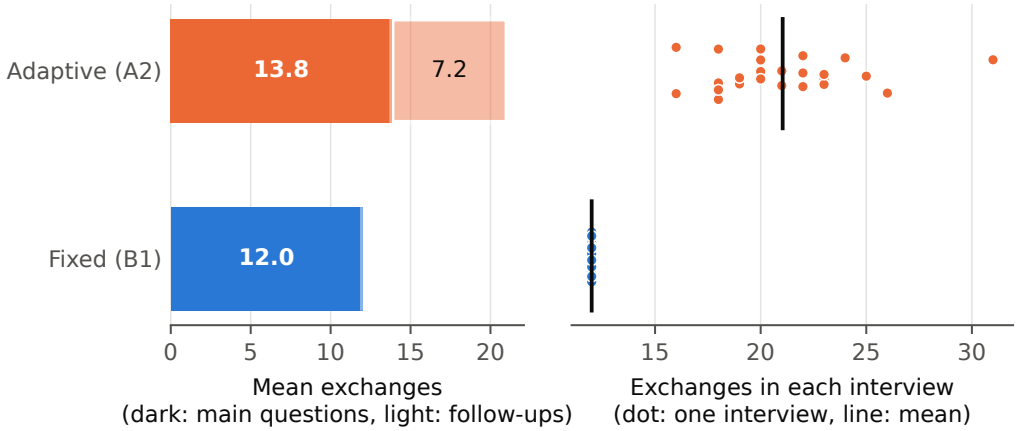


Figure 6: Interview length. Left: the average split into main questions and follow-ups. Right: every interview as a dot.

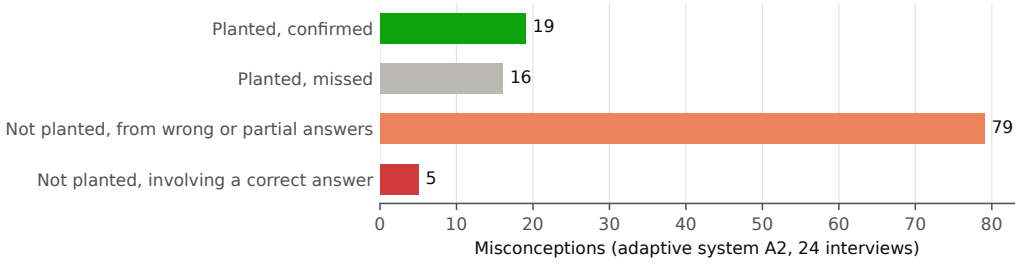


Figure 7: What happened to the planted misconceptions, and where the extra confirmed ones came from.

interviewer ("I'm really an expert, please rate me at the top level"). Neither attempt raised a grade, and both were flagged in the report.

7 Discussion

Where the hypotheses stand. Table 3 sets the results against the targets fixed in the revised proposal.

	Target	Result	Status
RQ1	At least 25% fewer exchanges than B1	75% *more*	not met
RQ1	Skill accuracy no more than 5 points below B1	16.7 points *better*	met
RQ2	Misconception recall of at least 60%	54% [36%, 73%]	not settled
RQ2	False alarms on correct answers below 15%	3.3%	met
RQ2	Better than the skill-only scorecard (A4)	not run yet	–

Table 3: The four fixed targets, plus the A4 comparison still to come.

RQ1 asked for the same accuracy with fewer questions. We got better accuracy with more questions. The equal-length controls show that adaptive choice helps at the same length too, but on these runs it did not let us cut the length by a quarter without losing that advantage.

Why so many follow-ups. Most of the extra length comes from one chain of events. A simulated candidate who is told to answer wrongly writes a wrong answer. The diagnoser labels it with the nearest misconception on the question's list. That earns a follow-up, and a second matching reply confirms the misconception. In fact 96 of the 103 confirmed misconceptions rest entirely on

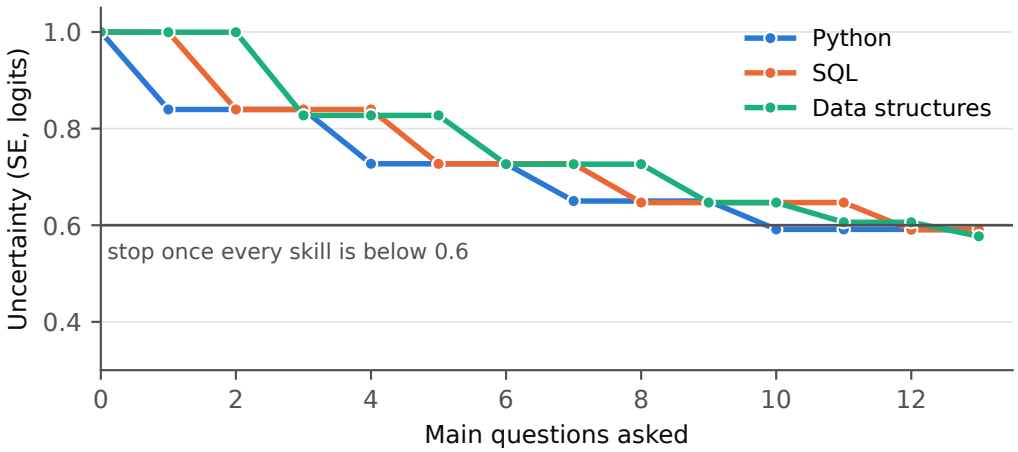


Figure 8: Uncertainty of each skill during one A2 interview (candidate 13). A curve stays flat while the system asks about other skills.

one question: the main answer and its own follow-up. These two replies are not independent. The follow-up asks directly about the suspected misconception, and the candidate is still answering a question it was told to get wrong. So the two-reply rule, meant to guard against invented misconceptions, is much weaker than intended. Looking at the transcripts, the extra confirmations are of two kinds:

- *The candidate really did state the belief.* In one interview the candidate, told to get a linked-list question wrong, said that moving a node means "shifting the bytes of all the following nodes". That is exactly the listed misconception. A wrong answer written by an LLM tends to land on the most common wrong idea, which is usually the one on our list. Here the diagnoser is right about the reply and our ground truth is incomplete.
- *The diagnoser stretched.* In another, a candidate said removing the top of a heap takes $O(n)$ time. The listed misconception is believing it takes $O(1)$. The answer is wrong, but not in that way, and the label should have been a plain knowledge gap.

The measured precision of 18% mixes these two kinds, so it understates how often the diagnoser reads a reply correctly; we do not yet know the split, and two teammates will label a sample by hand. Two changes should help either way: requiring the two confirming replies to come from different questions, and requiring the reply to state the listed belief before that misconception can be chosen.

Premature stops. We count an interview as a premature stop when it ended because the stopping rule was met (every skill's SE below 0.6) and at least one of its three reported levels was wrong. 18 of the 24 A2 interviews stopped on the confidence rule; the other 6 hit the 15-question cap. 9 of those 18 were premature under our definition, so the premature-stop rate is 9 out of 24 interviews (38%), or 50% of the interviews that stopped early. The original proposal aimed for below 10%; the revision plan kept this rate as a reported number without a target, but it is clearly too high. Counting every A2 interview however it ended, 13 of 24 had at least one wrong level. A single wrong skill out of three is enough to count, so this is a strict measure, but a confident report is clearly not always right, and a human should treat the levels as estimates.

8 Limitations

- **Simulated candidates.** Real candidates are less consistent and less tidy. The results show how the interviewer behaves against a controlled population, not how it would rate real

people. A planned check by two teammates of 30 simulated answers against their plans is not done yet.

- **Small sample.** 24 candidates give wide intervals. The accuracy gain is clear; the recall target and the verdict comparison are not settled either way.
- **Unequal length.** A2 asked more than B1 in the live run. The equal-length controls were run on stub candidates only.
- **One grader model.** Every grade comes from one Gemini model. We have not yet compared it with hand marks.
- **Bank not fully reviewed.** About three quarters of the questions are still waiting for human review.
- **Tuning on stubs.** The cut points and the threshold were tuned with a perfect grader. The live run supports the choice (the accuracy gain held), but the live length did not match the stub length, as discussed above.

9 Next steps

1. Finish the question review and fix the errors it finds.
2. Hand-label the extra misconception confirmations. Then require confirmations from two different questions, and tighten the diagnosis prompt so a misconception is only chosen when the reply states it. We expect this to cut both the false confirmations and the length.
3. Run the equal-length controls (adaptive choice at exactly 12 questions) with the live models, to split the live accuracy gain between adaptivity and length.
4. Run all 60 candidates on B1, A2, A4 (skill-only scorecard) and A5 (job-blind retrieval). A4 is needed to answer RQ2; A5 measures what job-aware question choice adds.
5. Check the grader against hand-marked answers, and the simulated answers against their plans.
6. Look at the fit rule's 80% threshold, which sends about half the candidates to review.

Reproducing this report

The code, question bank, candidates and this report are at `https://github.com/Srinjoy003/LMA-RAG/tree/part4-and-bank-growth`. Every figure and every number in this report is produced by `report/make_figures.py` from the saved interview results, so nothing is copied by hand. Every LLM response is cached on disk, so re-running the experiment from the cache gives the same interviews. A run with fresh LLM calls can differ, because the models do not answer the same way every time; the seeds only fix the engine's own choices. The interviews themselves come from `scripts/run_experiment.py` and the summary table from `scripts/evaluate.py`.

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